

Medical Record Science I (BBA(HM) 202)

All Modules Summary — 2024-2025 Sem 2

MODULE 1 — Introduction to Medical Records

Definition

- A medical record is a systematic, documented account of a patient's medical history, examinations, diagnoses, treatments, and outcomes.
- Full form: MR = Medical Record; MRD = Medical Records Department.

Importance of Medical Records

- Continuity of patient care, legal evidence, research, education, billing, quality assurance.

Characteristics of a Good Medical Record

- Accurate, complete, legible, timely, confidential, retrievable, and standardized.

Types of Medical Records

- Inpatient, Outpatient (OPD), Emergency, Diagnostic, Surgical, Medico-Legal, Administrative.

MODULE 2 — Organisation & Management of MRD

Organizational Structure

- Medical Record Officer → MR Technicians → MR Assistants → Coding/Statistics/MLC sections.

Functions of MRD

- Registration, record creation, assembling, deficiency checking, coding, indexing, filing, retrieval, statistics, birth/death registration.

Staffing Guidelines

- Based on bed strength, daily patient load, type of hospital, degree of computerization.

- General ratio: 1 MR technician per 50–100 beds.

Role of MR Technician

- Coding diagnoses (ICD), assembling records, deficiency checking, filing, statistics, medico-legal documentation.

MODULE 3 — Assembling, Indexing & Filing

Assembling

- Collecting all loose documents after discharge and arranging them in a standard order.
- Standard order: Face sheet → History & Physical → Progress notes → Orders → Lab reports → Radiology → Consent forms → Discharge summary.

Deficiency Checking

- Checking each record for missing signatures, diagnoses, reports, and consent forms.
- Deficiency slips sent to doctors for completion before filing.

Indexing

- **Disease Index:** Index by ICD diagnosis code.
- **Procedure Index:** Index by surgical/medical procedures performed.
- **Physician Index:** Index by treating doctor.
- **Patient Index (MPI):** Master Patient Index — unique identifier for each patient.

Filing Systems

- **Serial filing:** Records filed in order of registration number.
- **Terminal digit filing:** Filed by last two digits — reduces congestion at one area.
- **Middle digit filing:** Filed by middle digits.
- **Alphabetical filing:** Filed by patient name — suitable for smaller hospitals.

MODULE 4 — Retention & Preservation of Medical Records

Retention

- Policy of keeping records for a specified period before destruction.

- Adults: generally 5–10 years; Minors: till age of majority + specified years; MLCs: 10+ years.

Preservation Methods — Manual

- Microfilming, Microfiching, Lamination, Acid-free folders, Controlled storage (temperature/humidity), Pest control.

Preservation Methods — Electronic

- EHR systems, Scanning & digitization, Cloud storage, RAID, Optical discs, Encryption, Automated backup.

Computerization in MRD

- Hospital Information System (HIS), Electronic Medical Records (EMR), Picture Archiving & Communication System (PACS) for radiology.
- Benefits: Faster retrieval, reduced storage space, improved accuracy, remote access.

MODULE 5 — Medical Coding (ICD & POMR)

ICD — International Classification of Diseases

- Standardized system by WHO for coding diseases, conditions, and procedures.
- Current version: ICD-10 (ICD-11 being adopted globally).
- Used for billing, statistics, research, and insurance claims.

POMR — Problem Oriented Medical Record

- System introduced by Dr. Lawrence Weed.
- Components: Database → Problem List → Initial Plan → Progress Notes (SOAP format).
- SOAP: Subjective, Objective, Assessment, Plan.
- Advantage: Organizes patient information around problems for better care coordination.

Medical Coding — Brief Guide

- Translating diagnoses and procedures into alphanumeric codes.
- Essential for accurate billing and insurance reimbursement.

MODULE 6 — Hospital Statistics

- **Bed Occupancy Rate (BOR):** $(\text{Occupied bed days} / \text{Available bed days}) \times 100$.
- **Average Length of Stay (ALOS):** $\text{Total inpatient days} / \text{Total discharges}$.
- **Gross Death Rate:** $(\text{Total deaths} / \text{Total discharges including deaths}) \times 100$.
- **Net Death Rate:** $\text{Deaths after 48 hours} / \text{Total discharges} \times 100$.
- **Bed Turnover Rate:** $\text{Total discharges} / \text{Average beds available}$.
- **OPD/IPD ratio:** Reflects hospital utilization patterns.

Uses of Hospital Statistics

- Hospital planning, resource allocation, quality assessment, government reporting, research.

MODULE 7 — Legal Aspects of Medical Records

Medico-Legal Cases (MLC)

- Cases with legal implications: accidents, assaults, poisoning, rape, suspicious deaths.
- MLC must be reported to police; strict documentation and confidentiality required.

Legal Documents in Medical Records

- Consent forms, MLC register, Post-mortem report, Wound certificate, Birth/Death certificates, Discharge against medical advice (DAMA) form.

Patient Rights

- Right to access own records, right to confidentiality, right to informed consent, right to correction of errors.

Confidentiality

- Patient information cannot be disclosed without consent except to treating team, court orders, or public health authorities.

MODULE 8 — Medical Audit

Definition

- Systematic review of medical care quality against established standards.

Types

- Prospective, Concurrent, Retrospective; Internal vs External audit.

Audit Process (Audit Cycle)

- Select topic → Set standards → Collect data → Compare → Identify gaps → Implement changes → Re-audit.

Importance

- Improves patient care quality, reduces errors, supports accreditation (NABH/JCI), ensures accountability.